MERN JWTs, Axios, Splitting Code, and Mongoose

These instructions are for Linux, specifically Ubuntu was used.

This assumes that you have already completed sections A and B

Split Code

In the cards directory create a file named api.js.
In server.js take the 3 API endpoints. Copy them, past them into api.js, and then comment them out.
Add the exports into api.js as follows:

```
require('express');
require('mongodb');

exports.setApp = function ( app, client )
{

        app.post('/api/addcard', async (req, res, next) =>
        {
          // incoming: userId, color
          // outgoing: error

          const { userId, card } = req.body;

          const newCard = {Card:card,UserId:userId};
          var error = '';

          try
          {
                const db = client.db('COP4331Cards');
                const result = db.collection('Cards').insertOne(newCard);
          }
          catch(e)
          {
                error = e.toString();
          }

          cardList.push( card );

          var ret = { error: error };
          res.status(200).json(ret);
        });

        app.post('/api/login', async (req, res, next) =>
        {
          // incoming: login, password
          // outgoing: id, firstName, lastName, error

         var error = '';
```

```
    const { login, password } = req.body;

    const db = client.db('COP4331Cards');
    const results = await db.collection('Users').find({Login:login,Password:password}).toArray();

    var id = -1;
    var fn = '';
    var ln = '';

    if( results.length > 0 )
    {
        id = results[0].UserID;
        fn = results[0].FirstName;
        ln = results[0].LastName;
    }

    var ret = { id:id, firstName:fn, lastName:ln, error:''};
    res.status(200).json(ret);
  });

  app.post('/api/searchcards', async (req, res, next) =>
  {
    // incoming: userId, search
    // outgoing: results[], error

    var error = '';

    const { userId, search } = req.body;

    var _search = search.trim();

    const db = client.db('COP4331Cards');
    const results = await db.collection('Cards').find({"Card":{$regex:_search+'.*', $options:'i'}}).toArray();

    var _ret = [];
    for( var i=0; i<results.length; i++ )
    {
        _ret.push( results[i].Card );
    }

    var ret = {results:_ret, error:error};
    res.status(200).json(ret);
  });

}
```

In server.js add the following (I put this right below the database connection code.):

```
var api = require('./api.js');
api.setApp( app, client );
```

Now remove the API code from server.js

Create Path.js in src/components

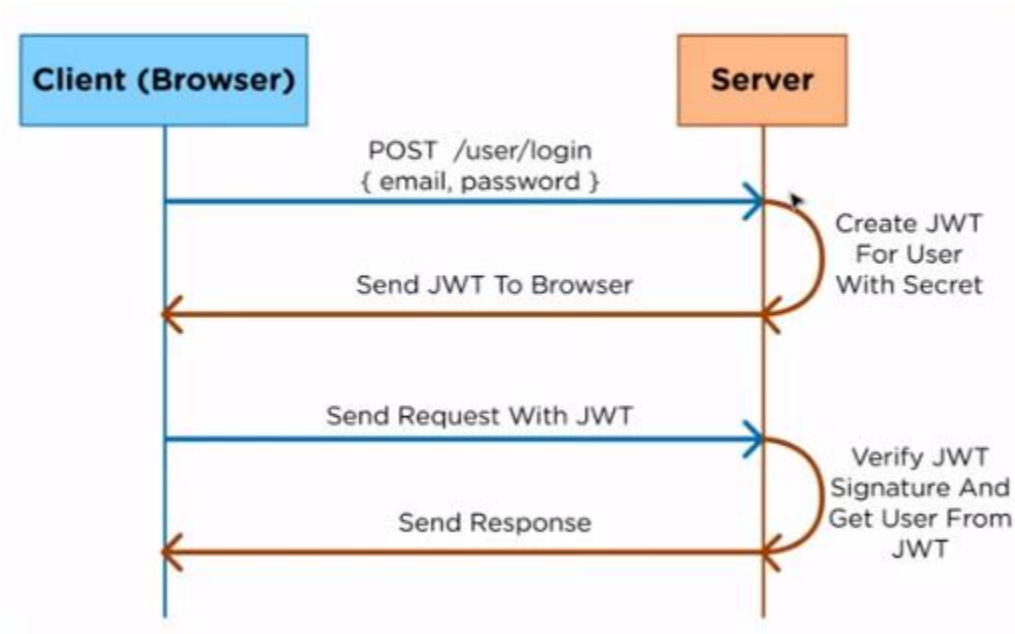
```
const app_name = 'cop4331-5.com'
exports.buildPath =
   function buildPath(route:string) : string
   {
      if (process.env.NODE_ENV != 'development')
      {
         return 'http://' + app_name + ':5000/' + route;
      }
      else
      {
         return 'http://localhost:5000/' + route;
      }
   }
```

Replace (in Login.tsx and CardUI.tsx) the calls to build path.

```
import { buildPath } from './Path';

const response = await fetch(buildPath('api/login'),
            {method:'POST',body:js,headers:{'Content-Type': 'application/json'}});
```

JWTs



From cmd line:

sudo npm install jsonwebtoken (in root and frontend)

sudo npm install dotenv ← you may have done this in section B

← also do this in the server directory and the

Frontend directory

Add an environmental variable named ACCESS_TOKEN_SECRET (make sure you do this for local in .env)

Create new file in cards directory named createJWT.js

```
const jwt = require("jsonwebtoken");
require("dotenv").config();

exports.createToken = function ( fn, ln, id )
{
    return _createToken( fn, ln, id );
}

_createToken = function ( fn, ln, id )
{
    try
    {
      const expiration = new Date();
      const user = {userId:id,firstName:fn,lastName:ln};

      const accessToken =  jwt.sign( user, process.env.ACCESS_TOKEN_SECRET);

      // In order to exoire with a value other than the default, use the
       // following
      /*
      const accessToken= jwt.sign(user,process.env.ACCESS_TOKEN_SECRET,
         { expiresIn: '30m'} );
                       '24h'
                      '365d'
      */

      var ret = {accessToken:accessToken};
    }
    catch(e)
    {
      var ret = {error:e.message};
    }
    return ret;
}

exports.isExpired = function( token )
{

   var isError = jwt.verify( token, process.env.ACCESS_TOKEN_SECRET,
     (err, verifiedJwt) =>
   {
     if( err )
     {
       return true;
```

```
      }
      else
      {
        return false;
      }
    });

    return isError;

}

exports.refresh = function( token )
{
  var ud = jwt.decode(token,{complete:true});

  var userId = ud.payload.id;
  var firstName = ud.payload.firstName;
  var lastName = ud.payload.lastName;

  return _createToken( firstName, lastName, userId );
}


exports.refresh = function( token )
{
  var ud = jwt.decode(token,{complete:true});

  var userId = ud.payload.id;
  var firstName = ud.payload.firstName;
  var lastName = ud.payload.lastName;

  return _createToken( firstName, lastName, userId );
}
```

From the cards directory: **sudo npm install jsonwebtoken**

Add to login api endpoint

```
    app.post('/api/login', async (req, res, next) =>
    {
      // incoming: login, password
      // outgoing: id, firstName, lastName, error

     var error = '';

      const { login, password } = req.body;

      const db = client.db('COP4331Cards');
```

```
      const results = await
db.collection('Users').find({Login:login,Password:password}).toArray
();

      var id = -1;
      var fn = '';
      var ln = '';

      var ret;

      if( results.length > 0 )
      {
        id = results[0].UserId;
        fn = results[0].FirstName;
        ln = results[0].LastName;

        try
        {
          const token = require("./createJWT.js");
          ret = token.createToken( fn, ln, id );
        }
        catch(e)
        {
          ret = {error:e.message};
        }
      }
      else
      {
          ret = {error:"Login/Password incorrect"};
      }

      res.status(200).json(ret);
    });
```

Add to addcard and searchcard api endpoints

(above)

```
var token = require('./createJWT.js');

    const { userId, card, jwtToken } = req.body;

    try
    {
     if( token.isExpired(jwtToken))
     {
```

```
      var r = {error:'The JWT is no longer valid', jwtToken: ''};
      res.status(200).json(r);
      return;
    }
  }
  catch(e)
  {
    console.log(e.message);
  }
```

Change the end of the methods

```
  var refreshedToken = null;
  try
  {
    refreshedToken = token.refresh(jwtToken);
  }
  catch(e)
  {
    console.log(e.message);
  }

  var ret = { error: error, jwtToken: refreshedToken };
  res.status(200).json(ret);
```

…and

```
  var refreshedToken = null;
  try
  {
    refreshedToken = token.refresh(jwtToken);
  }
  catch(e)
  {
    console.log(e.message);
  }

  var ret = { results:_ret, error: error, jwtToken: refreshedToken };

app.post('/api/addcard', async (req, res, next) =>
{
  // incoming: userId, color
  // outgoing: error
```

```
  const { userId, card, jwtToken } = req.body;

  try
  {
    if( token.isExpired(jwtToken))
    {
      var r = {error:'The JWT is no longer valid', jwtToken: ''};
      res.status(200).json(r);
      return;
    }
  }
  catch(e)
  {
    console.log(e.message);
  }

  const newCard = {Card:card,UserId:userId};
  var error = '';

  try
  {
    const db = client.db();
    const result = db.collection('Cards').insertOne(newCard);
  }
  catch(e)
  {
    error = e.toString();
  }

  var refreshedToken = null;
  try
  {
    refreshedToken = token.refresh(jwtToken);
  }
  catch(e)
  {
    console.log(e.message);
  }

  var ret = { error: error, jwtToken: refreshedToken };

  res.status(200).json(ret);
});
```

```
app.post('/api/searchcards', async (req, res, next) =>
{
  // incoming: userId, search
  // outgoing: results[], error

  var error = '';

  const { userId, search, jwtToken } = req.body;

  try
  {
    if( token.isExpired(jwtToken))
    {
      var r = {error:'The JWT is no longer valid', jwtToken: ''};
      res.status(200).json(r);
      return;
    }
  }
  catch(e)
  {
    console.log(e.message);
  }

  var _search = search.trim();

  const db = client.db();
  const results = await db.collection('Cards').find({"Card":{$regex:_search+'.*',
$options:'i'}}).toArray();

  var _ret = [];
  for( var i=0; i<results.length; i++ )
  {
    _ret.push( results[i].Card );
  }

  var refreshedToken = null;
  try
  {
    refreshedToken = token.refresh(jwtToken);
  }
  catch(e)
  {
    console.log(e.message);
  }
```

```
    var ret = { results:_ret, error: error, jwtToken: refreshedToken };

    res.status(200).json(ret);
  });
```

In the src directory add a file names tokenStorage.js

```
export function storeToken( tok:any ) : any
{
    try
    {
      localStorage.setItem('token_data', tok.accessToken);
    }
    catch(e)
    {
      console.log(e);
    }
}

export function retrieveToken() : any
{
    var ud;
    try
    {
      ud = localStorage.getItem('token_data');
    }
    catch(e)
    {
      console.log(e);
    }
    return ud;
}
```

Install react-jwt – sudo npm install react-jwt

CardUI.tsx

```
import React, { useState } from 'react';
import { buildPath } from './Path';
import { retrieveToken, storeToken } from '../tokenStorage';

function CardUI()
{
```

```
    const [message,setMessage] = useState('');
    const [searchResults,setResults] = useState('');
    const [cardList,setCardList] = useState('');
    const [search,setSearchValue] = React.useState('');
    const [card,setCardNameValue] = React.useState('');

    var _ud = localStorage.getItem('user_data');
    var ud = JSON.parse(String(_ud));
    var userId = ud.id;
//    var firstName = ud.firstName;
//    var lastName = ud.lastName;

    async function addCard(e:any) : Promise<void>
    {
        e.preventDefault();

        var obj = {userId:userId,card:card,jwtToken:retrieveToken()};
        var js = JSON.stringify(obj);

        try
        {
            const response = await fetch(buildPath('api/addcard'),
            {method:'POST',body:js,headers:{'Content-Type': 'application/json'}});

            let txt = await response.text();
            let res = JSON.parse(txt);

            if( res.error.length > 0 )
            {
                setMessage( "API Error:" + res.error );
            }
            else
            {
                setMessage('Card has been added');
                storeToken( res.jwtToken );
            }
        }
        catch(error:any)
        {
            setMessage(error.toString());
        }
    };

    async function searchCard(e:any) : Promise<void>
```

```
{
   e.preventDefault();

   var obj = {userId:userId,search:search,jwtToken:retrieveToken()};
   var js = JSON.stringify(obj);

   try
   {
      const response = await fetch(buildPath('api/searchcards'),
      {method:'POST',body:js,headers:{'Content-Type': 'application/json'}});

      let txt = await response.text();
      let res = JSON.parse(txt);
      let _results = res.results;
      let resultText = '';
      for( let i=0; i<_results.length; i++ )
      {
         resultText += _results[i];
         if( i < _results.length - 1 )
         {
            resultText += ', ';
         }
      }
      setResults('Card(s) have been retrieved');
      storeToken( res.jwtToken );
      setCardList(resultText);
   }
   catch(error:any)
   {
      alert(error.toString());
      setResults(error.toString());
   }
};

function handleSearchTextChange( e: any ) : void
{
   setSearchValue( e.target.value );
}

function handleCardTextChange( e: any ) : void
{
   setCardNameValue( e.target.value );
}
```

```
   return(
<div id="cardUIDiv">
  <br />
  Search: <input type="text" id="searchText" placeholder="Card To Search For"
    onChange={handleSearchTextChange} />
  <button type="button" id="searchCardButton" className="buttons"
    onClick={searchCard}> Search Card</button><br />
  <span id="cardSearchResult">{searchResults}</span>
  <p id="cardList">{cardList}</p><br /><br />
  Add: <input type="text" id="cardText" placeholder="Card To Add"
    onChange={handleCardTextChange} />
  <button type="button" id="addCardButton" className="buttons"
    onClick={addCard}> Add Card </button><br />
  <span id="cardAddResult">{message}</span>
</div>
   );
}

export default CardUI;
```

Login.tsx

```
import React, { useState } from 'react';
import { buildPath } from './Path';
import { storeToken } from '../tokenStorage';
import { jwtDecode } from 'jwt-decode';

function Login()
{

  const [message,setMessage] = useState('');
  const [loginName,setLoginName] = React.useState('');
  const [loginPassword,setPassword] = React.useState('');

    async function doLogin(event:any) : Promise<void>
    {
        event.preventDefault();

        var obj = {login:loginName,password:loginPassword};
        var js = JSON.stringify(obj);

        try
        {
            const response = await fetch(buildPath('api/login'),
```

```
        {method:'POST',body:js,headers:{'Content-Type': 'application/json'}});

      var res = JSON.parse(await response.text());

    const { accessToken } = res;
    storeToken( res );

    const decoded = jwtDecode(accessToken);

    try
    {
      var ud = decoded;
      var userId = ud.iat;
      var firstName = ud.firstName;
      var lastName = ud.lastName;

      if( userId <= 0 )
      {
        setMessage('User/Password combination incorrect');
      }
      else
      {
        var user = {firstName:firstName,lastName:lastName,id:userId}
        localStorage.setItem('user_data', JSON.stringify(user));

        setMessage('');
        window.location.href = '/cards';
      }
      }
      catch(e)
      {
        console.log( e );
        return;
      }
    }
    catch(error:any)
    {
        alert(error.toString());
        return;
    }
  };

function handleSetLoginName( e: any ) : void
{
```

```
    setLoginName( e.target.value );
  }

  function handleSetPassword( e: any ) : void
  {
    setPassword( e.target.value );
  }

  return(
    <div id="loginDiv">
      <span id="inner-title">PLEASE LOG IN</span><br />
      Login: <input type="text" id="loginName" placeholder="Username"
        onChange={handleSetLoginName} /><br />
      Password: <input type="password" id="loginPassword" placeholder="Password"
        onChange={handleSetPassword} />
      <input type="submit" id="loginButton" className="buttons" value = "Do It"
        onClick={doLogin} />
      <span id="loginResult">{message}</span>
   </div>
  );
};

export default Login;
```

Axios – Optional

From command line in the frontend directory use:
npm install axios

In Login.js and CardUI.js add the following:

```
import axios from 'axios'
```

Replace all network calls as follows:

bp.buildPath(‘api/login’)
bp.buildPath(‘api/addcard’)
bp.buildPath(‘api/searchcards’)

Remember: var storage = require('../tokenStorage.js');

```
import React, { useState } from 'react';
import axios from 'axios';

function Login()
```

```
{

    var bp = require('./Path.js');
    var storage = require('../tokenStorage.js');

    var loginName;
    var loginPassword;

    const [message,setMessage] = useState('');

    const doLogin = async event =>
    {
        event.preventDefault();

        var obj = {login:loginName.value,password:loginPassword.value};
        var js = JSON.stringify(obj);

        var config =
        {
            method: 'post',
            url: bp.buildPath('api/login'),
            headers:
            {
                'Content-Type': 'application/json'
            },
            data: js
        };

        axios(config)
            .then(function (response)
        {
            var res = response.data;
            if (res.error)
            {
                setMessage('User/Password combination incorrect');
            }
            else
            {
                storage.storeToken(res);
                var jwt = require('jsonwebtoken');

                var ud = jwt.decode(storage.retrieveToken(),{complete:true});
                var userId = ud.payload.userId;
                var firstName = ud.payload.firstName;
```

```
            var lastName = ud.payload.lastName;

            var user = {firstName:firstName,lastName:lastName,id:userId}
            localStorage.setItem('user_data', JSON.stringify(user));
            window.location.href = '/cards';
        }
      })
      .catch(function (error)
      {
        console.log(error);
      });
    }

    return(
      <div id="loginDiv">
        <span id="inner-title">PLEASE LOG IN</span><br />
        <input type="text" id="loginName" placeholder="Username" ref={(c) => loginName = c}  /><br
/>
        <input type="password" id="loginPassword" placeholder="Password" ref={(c) =>
loginPassword = c} /><br />
        <input type="submit" id="loginButton" class="buttons" value = "Do It"
          onClick={doLogin} />
        <span id="loginResult">{message}</span>
      </div>
    );
};

export default Login;
```

Mongoose – Optional

From command line use:
npm install mongoose, npm install mongoose-int32

Change the connection code

From

```
const MongoClient = require('mongodb').MongoClient;
const url = process.env.MONGODB_URI;
const client = new MongoClient(url);
client.connect();
```

To:

```
const url = process.env.MONGODB_URI;
const mongoose = require("mongoose");
mongoose.connect(url)
  .then(() => console.log("Mongo DB connected"))
  .catch(err => console.log(err));
```

Replace

```
api.setApp( app, client );
```

with

```
api.setApp( app, mongoose );
```

Create a models folder in the cards directory. Inside create user.js and card.js

user.js

```
const mongoose = require("mongoose");
const Schema = mongoose.Schema;
//Create Schema
const UserSchema = new Schema({
  UserId: {
    type: Number
  },
  FirstName: {
    type: String,
    required: true
  },
  LastName: {
    type: String,
    required: true
  },
  Login: {
    type: String,
    required: true
  },
  Password: {
    type: String,
    required: true
  }
});
module.exports = user = mongoose.model("Users", UserSchema);
```

card.js

```
const mongoose = require('mongoose');
const Schema = mongoose.Schema;
// Create Schema
const CardSchema = new Schema({
  UserId: {
    type: Number
  },
  Card: {
    type: String,
    required: true
  }
});

module.exports = Card = mongoose.model('Cards', CardSchema);
```

In api.js

```
//load user model
const User = require("./models/user.js");
//load card model
const Card = require("./models/card.js");
```

Change the database code to the following:

addcard

```
//const newCard = { Card: card, UserId: userId };
  const newCard = new Card({ Card: card, UserId: userId });
  var error = '';
  try
  {
    // const db = client.db();
    // const result = db.collection('Cards').insertOne(newCard);
    newCard.save();
  }
  catch (e)
  {
    error = e.toString();
  }
```

login

```
const { login, password } = req.body;
  // const db = client.db();
  // const results = await db.collection('Users').find({Login:login,Password:password}).toArray();
```

```
  const results = await User.find({ Login: login, Password: password });
```

searchcard

```
  var _search = search.trim();
  //   const db = client.db();
  //   const results = await db.collection('Cards').find({ "Card": { $regex: _search + '.*', $options: 'r' } }).toArray();
  const results = await Card.find({ "Card": { $regex: _search + '.*', $options: 'r' } });
```

In CardUI.js change line:

```
var userId = ud. id;
```

to the following:

```
var userId = ud. userId;
```